TEACHER'S NOTES

Maintenance of IO-Link Systems: Smart Sensors

Model IM4330 · CP3723 & CP7395

Field	Value
Issue	1.0
Date	2026-06-26
Supplier	Matrix Technology Solutions Ltd
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⚠ **WARNING** These notes are for the teacher or trainer only. Do not distribute to students.

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1. Standards Alignment

1.1 T-Level — Engineering, Manufacturing & Processing

Pathway: Maintenance, Installation & Repair of Engineering Systems

IM4330 Learning Activity	T-Level Core Component
IO-Link communication, MQTT, ISDU parameter access	K1 — Engineering principles and systems theory
Sensor commissioning, setpoint and calibration write	K2 — Installation and commissioning of systems
Fault scenario diagnosis and corrective ISDU write	K3 — Maintenance, inspection and fault diagnosis
HMI dashboard operation, live data interpretation	K4 — Digital tools and monitoring systems
CP7395 WS4 — Benefits for Maintenance cost study	K5 — Commercial and business awareness
CP7395 WS7 — PLC and HMI Integration	K6 — Control systems and automation
CP7395 WS8 — Standard vs IO-Link: The Numbers	S1 — Problem-solving using data and analysis
All fault scenarios (observe → diagnose → fix → verify)	S2 — Systematic fault diagnosis methodology

1.2 NVQ / Level 3 Diploma in Engineering Maintenance

Unit	IM4330 Coverage
Maintaining and Restoring Mechanical Systems	CP3723 WS7 — Fault Finding and Replacement (sensor swap procedure)
Maintaining Electrical/Electronic Equipment	CP3723 WS3–WS6 — sensor commissioning and ISDU configuration
Carrying Out Planned Maintenance	CP3723 WS5 — calibration offset scenario; maintenance schedule
Diagnosing and Correcting Faults	CP3723 WS7, WS8 — end-to-end fault diagnosis assessment
Using and Interpreting Engineering Data	CP7395 WS5 — PDin hex decode; WS10 — PT100 resistance curve

1.3 BTEC Level 3 National Extended Certificate in Engineering

Unit	IM4330 Coverage
Unit 1 — Engineering Principles	Sensor operating principles (CP3723 WS2–WS6)
Unit 3 — Engineering Product Design	System architecture (CP7395 WS1, WS2)
Unit 7 — Calculus to Solve Engineering Problems	4–20 mA calculations (CP3723 WS2 Q2), hex decode (CP7395 WS5, WS10)
Unit 19 — Electronic Devices and Circuits	IO-Link physical layer, PDin bit structure (CP7395 WS3)
Unit 21 — Electronic Measurement and Testing	Live ISDU read, connection diagnostics latency (CP7395 WS6)

1.4 IEC 61131-9 — IO-Link Standard

Standard Concept	Where Covered
Cyclic process data (PDin/PDout)	CP3723 WS3–WS6; CP7395 WS3, WS5
Acyclic service data (ISDU)	CP3723 WS3–WS6; CP7395 WS3

Event data (fault/warning flags)	CP7395 WS6
Device identity (Vendor ID / Device ID)	CP7395 WS9
IODD (IO-Link Device Description)	CP7395 WS3, WS9
COM2/COM3 speed classes	CP7395 WS1

1.5 Industry 4.0 Alignment

I4.0 Pillar	CP7395 Coverage
Industrial IoT	WS1, WS2 — MQTT, WebSocket, Pi as edge device
Big Data / Analytics	WS5, WS10 — data decode; WS6 — diagnostics data
Smart Sensors	WS3, WS9 — protocol-level sensor operation
Predictive Maintenance	WS4, WS8 — maintenance economics and event data
Connectivity	WS7 — PLC/HMI integration; WS2 — network architecture

2. Teaching Strategy

2.1 Three Pedagogical Modes

Mode	Description	Where Used
Guided	Teacher or worksheet leads step-by-step with a defined outcome	WS1 (kit intro), WS3–WS6 (sensor labs)
Fault-Led	A real fault is injected; student must diagnose and fix without knowing the answer	WS3 NC logic, WS5 calibration, WS7 maintenance
Discovery-Based	Open-ended exploration; student observes and draws conclusions	WS1 challenge, WS2 animations, WS4 counter task

2.2 Approach by Worksheet Type

Introductory worksheets (WS1–WS2) Let the hands-on challenge come first — do not explain everything upfront. The MCQ at the end reveals gaps; address them then.

Sensor labs (WS3–WS6) Each follows: operating principle → live observation → fault scenario. Do not preview the fault or hint at the diagnosis — genuine uncertainty is the point. The debug console and ISDU reads provide all needed information.

Fault finding (WS7) The light stack is the first clue before the student looks at the sensor — mirroring real industrial practice. Encourage students to work through the CL50 state before opening the IO-Link Master page.

Final assessment (WS8) Individual assessment. No teacher support during Tasks 2A, 3, and 4. Task 1 (device identification) may be supported.

CP7395 engineering worksheets Best delivered after CP3723 is complete. WS1–WS4 can run in parallel with CP3723 WS3–WS6 for mixed-ability groups. Check hexadecimal prerequisite before assigning WS5 independently.

2.3 Grouping and Differentiation

- Pairs recommended for CP3723 — one operates the HMI, one handles sensors; swap for each worksheet.
- Individual work for CP7395 and WS8 (final assessment).
- Extension: direct fast-finishing students to the Admin / Diagnostics latency graph, or to the IO-Link Master IODD parameter cards to decode raw hex manually.
- Support: use the Progress page to identify students who have not visited certain worksheets.

2.4 Common Misconceptions

Misconception	Correction
"IO-Link replaces OUT1"	IO-Link runs alongside OUT1 on the same cable — the switching output still works normally. (WS2 Q3)
"The mA value equals the engineering value"	4–20 mA is a scaled range — 12 mA is the midpoint, not 12 bar / 12 °C. (WS2 Q2)
"NC means the sensor is broken"	NC (Normal Closed) is a valid configuration — output ON when nothing is present. It must be intentional. (WS3 scenario, WS7)
"Replace the sensor if the reading drifts"	Calibration offset (ISDU index 681) corrects drift in the field without replacement. (WS5 scenario, WS8 Q3)

"The instability alarm means the sensor has failed"	Instability means the target is near the edge of sensing range — reposition before it causes a fault. (WS3 Q2, WS8 Q1)
"500 ms publish interval = 500 messages per second"	500 ms is the period — that is 2 messages per second. (CP7395 WS1 Q2)

3. Suggested Delivery Schedule

3.1 CP3723 — Maintenance on Smart Sensors

Recommended: 8–10 hours total across 4–5 weekly 2-hour sessions.

Session	Worksheets	Time	Notes
1	WS1 — Meet Your Kit	60–75 min	Allow extra time for the capacitive challenge. Debrief MCQ as a class.
1	WS2 — What is a Smart Sensor?	45–60 min	Watch the three signal animations together before students attempt the fault scenario.
2	WS3 — The Proximity Sensor	60–75 min	The NC scenario is the first real fault injection — give students time to be confused before intervening.
2	WS4 — The Capacitive Sensor	45–60 min	The detection counter is a quick win — confirms ISDU access is working.
3	WS5 — The Temperature Sensor	75–90 min	Longest worksheet. The calibration scenario requires a 3-second verify hold — allow time.
3	WS6 — The Light Stack	60 min	Students enjoy colour control. Use to explain PDout before the hex decode in WS8.
4	WS7 — Fault Finding	60–75 min	Semi-independent. Resist prompting the diagnosis.
4/5	WS8 — Final Assessment	45–60 min	Individual. Tasks 2A, 3, and 4 are unaided.

3.2 CP7395 — Industry 4.0 IO-Link

Recommended: 10–12 hours total across 5–6 weekly 2-hour sessions.

Session	Worksheets	Time	Notes
1	WS1 — Your System: A Technical Brief	45 min	Good revision of CP3723 hardware from a technical perspective.
1	WS2 — How This App Works	60 min	The Admin / Diagnostics latency graph is a useful live demo.
2	WS3 — How Each Sensor Works (Protocol)	75 min	Requires hexadecimal knowledge. Pre-check or introduce hex first.
2	WS4 — Benefits for Maintenance	45 min	The process / event / service dropdown is the key activity.
3	WS5 — Decoding IO-Link Data	60 min	Work through the first decode exercise together before students attempt independently.
3	WS6 — Diagnostics: Process, Service, Event	45 min	Use alongside the live Admin page.
4	WS7 — PLC and HMI Integration	60 min	No PLC required — concepts illustrated via HMI data flow.
4	WS8 — Case Study: The Numbers	45 min	Work through the financial calculation (18 h × £5,000) on the board first.
5	WS9 — Device Identity	45 min	Vendor ID lookup on the IO-Link Master page is a live activity.
5	WS10 — PT100 Temperature Sensors	60 min	Two's complement exercises need care — work the −10.0 °C example as a class.

6	WS11 — CL50 PDout Encoding	60 min	The encode exercises are the most challenging task in CP7395.

4. Safety Briefing

Deliver this briefing before the first session (~5 minutes). Script below can be read verbatim.

Voltages

> *"The kit runs at 24 volts DC — the same voltage used on real factory floors. About the same as a car battery. It will not give you a dangerous shock under normal use, but treat it with respect. The Raspberry Pi also runs from this supply — an internal DC-to-DC converter on the panel steps the 24 volts down to 5 volts for it."*
>
> *"There is no mains voltage anywhere on this panel. Do not connect anything to a mains socket other than the supplied power adapter."*

Cables and Connectors

> *"The sensors use M12 connectors — round connectors with a locking ring. If you need to swap a sensor, turn off the 24-volt supply first, then twist the locking ring anticlockwise and pull. Don't yank the cable."*
>
> *"If you see a damaged cable — cut jacket, bent connector, anything like that — tell me before using it."*

Fault Scenarios

> *"Some worksheets will ask you to start a fault scenario. This writes a software change to the sensor over IO-Link — no wiring changes. The sensor will behave differently — that's the point — but it is completely safe. When you navigate away, the sensor restores itself automatically."*
>
> *"If for any reason the sensor isn't restoring, tell me. Don't try to fix it by unplugging sensors while the power is on."*

Light Stack and Infrared

> *"The Banner light stack contains bright LEDs. Don't stare directly into it at close range, especially on High intensity."*
>
> *"The photoelectric sensor emits an infrared beam — don't look directly into the sensor face."*

If Something Goes Wrong

> *"If you smell burning, see smoke, or anything looks wrong — don't touch anything. Tell me. The only thing you should do yourself is unplug the power supply from the wall, which cuts all power immediately."*

Pre-Session Checklist (Trainer)

- ☐ All M12 cable connectors are seated and locked
- ☐ No cables are kinked, pinched, or damaged
- ☐ The power supply is on and the HMI kiosk has loaded
- ☐ IO-Link Master page shows all five devices connected
- ☐ Temperature sensor (Port 4) shows a plausible ambient reading (not −40 °C)
- ☐ CL50 light stack shows steady green (default startup state)

5. Worksheet Answers — CP3723

WS1 — Meet Your Kit

Question	**Answer**	Explanation
Q1	**A**	The IO-Link Master is orange and is sometimes called the hub.
Q2	**C**	The proximity sensor on Port 1 detects metal using an electromagnetic field.
Q3	**B**	The capacitive sensor detects liquids and powders through a container wall.
Q4	**A**	The Raspberry Pi collects sensor data, runs the communications, and serves the dashboard.
Q5	**C**	The capacitive sensor showed a live rising level before fully triggering — not just on/off.

WS2 — What is a Smart Sensor?

Question	**Answer**	Explanation
Q1	**A**	A basic digital sensor goes silent on failure — no diagnostic information available.
Q2	**C**	12 mA is the midpoint of the 4–20 mA range. 8 mA span ÷ 2 = 4 mA above floor = 50% → 50 bar.
Q3	**A**	IO-Link carries process data, fault codes, device identity, and remote parameter writes over the same cable without interrupting OUT1.
Q4	**C**	An analogue signal gives a continuous level reading, not just full/empty.
Q5	**B**	The square wave represents OUT1 — still present and working while the IO-Link data stream runs above it.

WS2 — Fault Diagnosis Scenario (randomised — 5 possible faults)

> ***Trainer note:*** *The fault injected changes on every "Start Scenario" click. Identify which fault ran by matching the amber [WARN] hints in the debug console.*

Fault injected	Correct answer option	Debug console hint
1 — Temp over SP1 (Port 3)	Option describing high temperature / SP1 exceeded	[WARN] Port3: Process value exceeds SP1
2 — Proximity OUT1 stuck HIGH (Port 1)	Option describing output stuck / no metal in field	[WARN] Port1: OUT1 asserted with no metal target
3 — Wrong device identity on Port 2	Option describing device identity mismatch	[WARN] Port2: Device identity mismatch
4 — IO-Link comms loss on Port 3	Option describing communication loss / port offline	[WARN] Port3: IO-Link connection dropped
5 — Master supply undervoltage	Option describing supply voltage below threshold	[WARN] Supervision: Supply voltage below threshold

WS3 — The Proximity Sensor

Question	**Answer**	Explanation
Q1	**A**	OUT1 jumps from 0 to 1 when metal is detected, drops back to 0 on removal.
Q2	**C**	Instability alarm = target near the edge of sensing range. Detection may be unreliable.

Q3	**A**	Increase the standoff distance between the sensor face and the target.

WS3 — NC Logic Scenario (step-by-step)

Step	Student action	Expected result
1 — Start	Click "Start Scenario"	OUT1 shows ON with nothing present (NC behaviour)
2 — Observe	Click "I have observed the anomaly"	Diagnosis options appear
3 — Diagnose	Select "Output logic changed from NO to NC"	ISDU read step unlocks
4 — Read	ISDU read: index 61 / sub 1	Returns value 1 (NC)
5 — Fix	ISDU write: index 61 / sub 1, value **0** (NO)	Verify step appears
6 — Verify	Hold metal target near sensor for 3 seconds	OUT1 ON with metal, OFF without — pass
7 — Sign-off	Tick all 3 checkboxes	Close button enables

Trainer note: *Auto-restore: navigating away at any step writes value 0 back to index 61/sub1 automatically.*

WS4 — The Capacitive Sensor

Question	**Answer**	Explanation
Q1	**C**	A new count triggers the moment detection goes OFF→ON. One touch = one count regardless of hold time.
Q2	**A**	Capacitive sensors detect water and powder through container walls — proximity sensors cannot.
Q3	**A**	Always-ON with no target = sensitivity too high, detecting the container wall. Lower SP1 via ISDU.

WS5 — The Temperature Sensor

Question	**Answer**	Explanation
Q1	**C**	See live temperature in °C, trend over time, and get early warning before the trip point is reached.
Q2	**B**	The alarm turns red and shows "ABOVE THRESHOLD" once temperature crosses the slider value.
Q3	**A**	−40 °C is the bottom of the TV7105 range and appears when the sensing element is open-circuit — broken or disconnected sensor.

WS5 — Calibration Drift Scenario CAL-0189 (step-by-step)

Step	Student action	Expected result
1 — Start	Click "Start Scenario"	Reading rises ~3 °C above ambient
2 — Observe	Click "I have observed the drift"	Diagnosis options appear
3 — Diagnose	Select "Write 0.0 °C — restore factory default"	ISDU read step unlocks
4 — Read	ISDU read: index 681 / sub 0	Returns value 3.0 (+3.0 °C offset)
5 — Fix	ISDU write: index 681 / sub 0, value **0.0**	Verify step appears

6 — Verify	Reading stays within ±1.5 °C of pre-inject baseline for 3 s	Pass
7 — Sign-off	Tick all 3 checkboxes	Close button enables

Trainer note: *Wrong diagnosis options: "Write +3.0 °C" and "Write −6.0 °C". Auto-restore: navigating away writes 0.0 back to index 681/sub0.*

Dynamic setpoints (calculated from live temperature at worksheet load):

Variable	Formula	Example at 27.3 °C
SP2 (OUT2 pre-warning)	ceil(temp) + 1	29 °C
SP1 (OUT1 main alarm)	ceil(temp) + 3	31 °C
RP1 (OUT1 reset)	ceil(temp)	28 °C
RP2 (OUT2 reset)	exact live temp	27.3 °C

WS6 — The Light Stack

Question	**Answer**	Explanation
Q1	**A**	PDout-only means the master sends a command TO the light — no measurement comes back.
Q2	**C**	Hex 180101: Octet2 = 0x01; low nibble = 1 = Red.
Q3	**A**	Flashing red held in the PLC output register — the fault state is in the controller, not the light.
Q4	**C**	Full colour/animation/intensity/speed over one 3-pin cable, plus HMI can read back exact state without a feedback wire.

WS6 — Maintenance Scenario LST-0312

✓ **ANSWER** Correct answer: C — PLC output register still holding 184203 (Amber Flash). Normal-run value 000100 (Green Steady) was never sent after PLC restart.

WS7 — Fault Finding and Replacement

Trainer note: *No MCQ in this worksheet — the scenario IS the assessment. The light stack colour is the first clue; encourage students to read it before opening the IO-Link Master page.*

Step	Student action	Expected result
1 — Observe CL50	Note light stack colour — fault indicator visible	First clue before touching sensor
2 — Check Master	IO-Link Master page — Port 1 OUT1 shows ON with no metal	Confirms sensor abnormality
3 — Diagnose	Select option **C** (NC logic)	NC logic identified as root cause
4 — Fix	ISDU write: index 61 / sub 1, value **0** (NO)	Sensor restored to normal operation

✓ **ANSWER** Correct diagnosis: C — output logic set to NC. OUT1 is ON when nothing is present and OFF when metal is detected — inverse of correct NO behaviour.

WS8 — Final Practical Assessment

Task 1 — Device Identification (observational — no fixed answer; student reads live IO-Link Master page):

Port	Device	Vendor ID
Port 1	Contrinex LTR-M18PA photoelectric (or Omron E2E proximity, depending on config)	342 or 612
Port 2	Carlo Gavazzi / RS PRO M18 capacitive	1586
Port 3	IFM TV7105 temperature	310
Port 4	Banner CL50 Pro light stack	451

Task 2A — Hex Decode (free-response, unaided):

✓ **ANSWER** Index 583 (SP1) returns 01 F4 (int16, ×0.1). Answer: 50.0 °C. Working: 0x01F4 = 500; 500 × 0.1 = 50.0

Task 3 — PDout Colour (free-response, unaided):

✓ **ANSWER** PDout 000109 → Octet2 = 0x09; low nibble = 9 → Colour index 9 = Blue. Answer: Blue

Task 4 — Knowledge Check MCQ (unaided):

Question	**Answer**	Explanation
Q1	**C**	Investigate — instability means target at edge of sensing range. Reposition before it causes a fault.
Q2	**A**	Output logic NC — write value 0 (NO) via ISDU index 61/sub1 to correct.
Q3	**B**	Write −3.0 °C calibration offset to index 681 via ISDU. Replacing the sensor is unnecessary.
Q4	**C**	Single 3-byte PDout controls all colour/animation/intensity over one cable vs. separate wire per segment conventionally.

6. Worksheet Answers — CP7395

WS1 — Your System: A Technical Brief

Question	**Answer**	Explanation
Q1	**A**	AL1350 IO-Link Master IP = 192.168.7.4.
Q2	**B**	500 ms interval = 2 messages per second. Common error: confusing period with frequency.
Q3	**C**	eth0 + wlan0 bridges the isolated IO-Link subnet to the building LAN.
Q4	**A**	All four ports (1–4) have a sensor connected. Port 5 has the CL50 light stack.

WS2 — How This App Works

Question	**Answer**	Explanation
Q1	**A**	MQTT publish/subscribe — AL1350 pushes data when ready without the Pi requesting each cycle.
Q2	**C**	ensure_mqtt_subscription() is called on every backend startup, re-registering the push subscription after AL1350 power-cycle.
Q3	**A**	FastAPI crash → live data stops and WebSocket disconnects.
Q4	**B**	High round-trip latency = AL1350 under load or congestion on the IO-Link subnet.

WS3 — How Each Sensor Works (Protocol Level)

Question	**Answer**	Explanation
Q1	**A**	PDin 0x01 — bit 0 set = object detected.
Q2	**C**	A running count tracks container fills without a separate counter sensor.
Q3	**C**	0x02EE = 750 decimal; 750 ÷ 10 = 75.0 °C.
Q4	**A**	PDout packs colour, animation, intensity, speed into specific bit positions across 3 bytes — one wrong bit changes everything.

WS4 — Benefits for Maintenance

Dropdown classification:

Scenario	**Correct Answer**
Temperature 23.5 °C arriving every 500 ms	**process**
Device replacement detected on Port 2	**event**
Parameter read — filter time = 10 ms	**service**
Object present flag (OUT1 switching state)	**process**
Short circuit fault reported	**event**

Question	**Answer**	Explanation
Q2	**A**	Process data + event data merged into a single JSON payload per WebSocket push.

WS5 — Decoding IO-Link Data

Hex	Type	Scale	**Answer**	Working
00 E6	int16	× 0.1	**23.0 °C** (RP1)	0x00E6 = 230; 230 × 0.1 = 23.0

00 64	uint16	none	**100 counts**	0x0064 = 100
01 F4	int16	× 0.1	**50.0 °C** (SP1)	0x01F4 = 500; 500 × 0.1 = 50.0

WS6 — Diagnostics: Process, Service, and Event Data

Data item	**Correct Answer**
Temperature 23.5 °C arriving every 500 ms	**process**
Device replacement detected on Port 2	**event**
Parameter read — filter time = 10 ms	**service**
Object present flag	**process**
Short circuit fault	**event**

WS7 — PLC and HMI Integration

Backend component matching:

Function	**Correct Answer**
Subscribing to AL1350 MQTT topics	**io_link_fastapi.py**
Decoding raw PDin into engineering values	**decoder.py**
Pushing decoded data to the browser	**io_link_fastapi.py**

Question	**Answer**	Explanation
Q1	**C**	IO-Link master maps each port's PDin into its process image; PLC reads this via fieldbus every scan cycle.
Q3	**C**	Faster fault ID, multi-machine monitoring, and historical trends without a site visit.
Q4	**A**	Spike in latency graph = high load on IO-Link subnet or AL1350 overloaded.

WS8 — Case Study: Standard vs IO-Link

Question	**Answer**	Working
Q1	**A** — 18 hours	Standard: 40 × 35 min = 1,400 min = 23.3 h. IO-Link: 40 × 8 min = 320 min = 5.3 h. Saving = 18 h.
Q2	**C** — £90,000	18 h × £5,000/h = £90,000
Q3	**B**	Automatic parameter restore on sensor swap + remote diagnostics = primary TCO drivers.
Q4	**A**	Event data (faults/warnings) triggers maintenance work orders in a CMMS.

WS9 — Device Identity: Vendor ID, Device ID & PDin

Question	**Answer**	Explanation
Q1	**A**	Port 1 (Omron E2E) reports Vendor ID 612.
Q2	**B**	Vendor ID 612 = OMRON Corporation.
Q3	**C**	Device ID uniquely identifies the exact model for ordering; master can restore parameters after swap.
Q4	**A**	PDin byte 1 bit 4 = instability alarm — check alignment, target at sensing range edge.
Q5	**C**	Vendor ID 1586, Device ID 1052673 = RS Pro M18 capacitive (Carlo Gavazzi OEM).

WS10 — PT100 Temperature Sensors

PDin Hex	**Answer**	Working
01 2C FF 00	**30.0 °C**	0x012C = 300; 300 ÷ 10
01 96 FF 00	**40.6 °C**	0x0196 = 406; 406 ÷ 10
FF 9C FF 00	**−10.0 °C**	0xFF9C = −100 (two's complement); ÷10
FE 70 00 00	**−40.0 °C**	0xFE70 = −400; ÷10 — open-circuit default

Trainer note: *Two's complement for FF9C: invert → 0063 = 99; add 1 = 100; apply sign → −100 × 0.1 = −10.0 °C*

WS11 — CL50 Pro PDout Encoding Deep-Dive

Decode exercises:

PDout Hex	**Decoded meaning**
000100	**Green · Steady · Medium speed · High intensity** — AL1350 startup default
004312	**Orange/Red · Two Colour Flash · Fast · High intensity**
198205	**Lime Green · Flashing · Slow · Low intensity**

Encode exercises:

Specification	Working	**Correct Hex**
Blue · Steady · Medium · High	Octet0: 00. Octet1: Speed=0, Anim=1 (Steady) → 01. Octet2: C1=9 (Blue) → 09.	**000109**
Cyan/Magenta · Two Colour Flash · Fast · High	Octet0: 00. Octet1: Speed=1 (Fast), Anim=3 (TwoColour) → 43. Octet2: C2=11 (Magenta), C1=7 (Cyan) → B7.	**0043B7**

7. Quick Reference Answer Key

Compact summary for use during live sessions. Full explanations in Sections 5 and 6.

CP3723 — MCQ Answers at a Glance

Worksheet	Q1	Q2	Q3	Q4	Q5
WS1 — Meet Your Kit	A	C	B	A	C
WS2 — What is a Smart Sensor?	A	C	A	C	B
WS3 — The Proximity Sensor	A	C	A	—	—
WS4 — The Capacitive Sensor	C	A	A	—	—
WS5 — The Temperature Sensor	C	B	A	—	—
WS6 — The Light Stack	A	C	A	C	—
WS7 — Fault Finding	scenario only	—	—	—	—
WS8 — Task 4 MCQ	C	A	B	C	—

CP3723 — Scenario Answers at a Glance

Worksheet	Correct diagnosis	Fix (ISDU write)
WS3 — NC Logic	Output logic changed from NO to NC	Index 61 / sub 1, value 0 (NO)
WS5 — Calibration Drift	Write 0.0 °C — restore factory default	Index 681 / sub 0, value 0.0
WS6 — LST-0312	C — PLC holding fault PDout value 184203	N/A — PLC issue, not sensor
WS7 — Fault Finding	C — Output logic set to NC	Index 61 / sub 1, value 0 (NO)
WS8 Task 2A	50.0 °C (0x01F4 = 500 × 0.1)	—
WS8 Task 3	Blue (Octet2 low nibble = 9)	—

CP7395 — MCQ Answers at a Glance

Worksheet	Q1	Q2	Q3	Q4	Q5
WS1 — Technical Brief	A	B	C	A	—
WS2 — How This App Works	A	C	A	B	—
WS3 — Sensor Protocol Level	A	C	C	A	—
WS4 — Benefits (Q2 only)	—	A	—	—	—
WS7 — PLC & HMI (Q1,3,4)	C	—	C	A	—
WS8 — Case Study	A	C	B	A	—
WS9 — Device Identity	A	B	C	A	C

CP7395 — Classification, Calculation & Decode at a Glance

Exercise	Answer
WS4 & WS6 — Temperature 23.5 °C every 500 ms	process
WS4 & WS6 — Device replacement on Port 2	event
WS4 & WS6 — Parameter read, filter time = 10 ms	service
WS4 & WS6 — Object present flag	process
WS4 & WS6 — Short circuit fault	event

WS5 — 00 E6 (int16 × 0.1)	23.0 °C
WS5 — 00 64 (uint16)	100 counts
WS5 — 01 F4 (int16 × 0.1)	50.0 °C
WS8 — Annual downtime saving	18 hours (23.3 h − 5.3 h)
WS8 — Annual financial saving	£90,000 (18 h × £5,000)
WS10 — 01 2C FF 00	30.0 °C
WS10 — 01 96 FF 00	40.6 °C
WS10 — FF 9C FF 00	−10.0 °C
WS10 — FE 70 00 00	−40.0 °C (open-circuit)
WS11 decode — 000100	Green · Steady · Medium · High
WS11 decode — 004312	Orange/Red · Two Colour Flash · Fast · High
WS11 decode — 198205	Lime Green · Flashing · Slow · Low
WS11 encode — Blue · Steady · Medium · High	000109
WS11 encode — Cyan/Magenta · Two Colour Flash · Fast · High	0043B7

End of Teacher's Notes — IM4330 Issue 1.0

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